MIVA
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COS 201

PROGRAMMING I



COS201 PROGRAMMING I: MID-SEMESTER LAB ASSESSMENT PROJECT

Case Study: Student Course Management (Mini-System)

Project Description

Your task is to design and implement a simple Student Course Management System. The purpose of the system is to help students record and manage the courses they are taking for the semester. The application should allow users to add courses, display all recorded courses, search for specific courses, compute total credit units, and save or load their course list from a file. You may implement the program using a text-based menu or a simple graphical interface.

PROJECT REQUIREMENTS

1. Core Functionality

- Add new courses with details such as course code, course title, and unit.
- Store all courses in a manageable internal structure.
- Display all stored courses clearly.
- Search for a course using its code.
- Calculate and display the total number of units entered.
- Save recorded courses to a file.
- Load previously saved courses from a file.

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2. Program Design

Your solution should demonstrate:

- Proper use of classes and objects.
- Methods for performing various operations.
- Logical decision-making for menu selection.
- Effective handling of invalid input.
- Simple string processing (e.g., trimming, casing, validation).
- Clean and structured coding style.

3. Additional Technical Features

Your program should include:

- Use of iteration (loops).
- Use of a simple recursive feature (e.g., a recurring menu display or a recursive search/count function).
- Appropriate error handling through exceptions.
- File reading and writing for saving/loading the course list.
- A simple GUI or an organized console-based interface.

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4. Expected Operations/Features

At minimum, your program should provide the following operations:

1. Add Course
2. View All Courses
3. Search Course by Code
4. Compute Total Units
5. Save to File
6. Load from File
7. Exit Program

SUBMISSION GUIDELINES

Each student must submit:

- All source code files.
- A screenshot showing the program running.
- A short report (1–2 pages) summarizing:

§ What the system does,

§ How each major feature works,

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§ Challenges encountered, i.e How errors were handled,

§ Where recursion was applied.

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